

Code:-

```
CREATE DATABASE IF NOT EXISTS employeeedb2;
USE employeeedb2;
```

```
CREATE TABLE IF NOT EXISTS departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50)
);
```

```
CREATE TABLE IF NOT EXISTS employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    dept_id INT,
    salary DECIMAL(10,2)
);
```

```
-- Insert Data
INSERT IGNORE INTO departments VALUES
(1,'CSE'),
(2,'ECE'),
(3,'EEE');
```

```
INSERT IGNORE INTO employees VALUES
(101,'Rahul',1,50000),
(102,'Priya',2,65000),
(103,'Anil',1,55000),
(104,'Sneha',3,70000),
(105,'Kiran',2,48000);
```

```
-- Verify the Tables
SHOW TABLES;
SELECT * FROM departments;
SELECT * FROM employees;
```

-- ===== -- 2. EXAMPLE VIEWS (1 - 14) -- =====

-- View 1: Display All Employees

```
DROP VIEW IF EXISTS EmployeeView;
CREATE VIEW EmployeeView AS
SELECT * FROM employees;
SELECT * FROM EmployeeView;
```

-- View 2: Selected Columns

```
DROP VIEW IF EXISTS EmployeeBasicView;
CREATE VIEW EmployeeBasicView AS
SELECT emp_id, emp_name, salary FROM employees;
SELECT * FROM EmployeeBasicView;
```

-- View 3: High Salary Employees

```
DROP VIEW IF EXISTS HighSalaryEmployees;
CREATE VIEW HighSalaryEmployees AS
SELECT emp_id, emp_name, salary FROM employees
WHERE salary > 55000;
SELECT * FROM HighSalaryEmployees;
```

-- View 4: Employee and Department using JOIN

```
DROP VIEW IF EXISTS EmployeeDepartmentView;  
CREATE VIEW EmployeeDepartmentView AS  
SELECT e.emp_id, e.emp_name, d.dept_name, e.salary  
FROM employees e  
JOIN departments d ON e.dept_id = d.dept_id;  
SELECT * FROM EmployeeDepartmentView;
```

-- View 5: Employees from a Particular Department

```
DROP VIEW IF EXISTS CSEmployees;  
CREATE VIEW CSEmployees AS  
SELECT e.emp_id, e.emp_name, d.dept_name, e.salary  
FROM employees e  
JOIN departments d ON e.dept_id = d.dept_id  
WHERE d.dept_name = 'CSE';  
SELECT * FROM CSEmployees;
```

-- View 6: Department-wise Average Salary

```
DROP VIEW IF EXISTS DepartmentSalaryView;  
CREATE VIEW DepartmentSalaryView AS  
SELECT d.dept_name, COUNT(e.emp_id) AS EmployeeCount, AVG(e.salary) AS AverageSalary  
FROM employees e  
JOIN departments d ON e.dept_id = d.dept_id  
GROUP BY d.dept_id, d.dept_name;  
SELECT * FROM DepartmentSalaryView;
```

-- View 7: Departments with Average Salary Above 55000

```
DROP VIEW IF EXISTS HighAverageDepartments;  
CREATE VIEW HighAverageDepartments AS  
SELECT d.dept_name, AVG(e.salary) AS AverageSalary  
FROM employees e  
JOIN departments d ON e.dept_id = d.dept_id  
GROUP BY d.dept_id, d.dept_name  
HAVING AVG(e.salary) > 55000;  
SELECT * FROM HighAverageDepartments;
```

-- View 8: Salary Range View

```
DROP VIEW IF EXISTS SalaryRangeView;  
CREATE VIEW SalaryRangeView AS  
SELECT emp_id, emp_name, salary FROM employees  
WHERE salary BETWEEN 50000 AND 70000;  
SELECT * FROM SalaryRangeView;
```

-- View 9: View with Calculated Column

```
DROP VIEW IF EXISTS AnnualSalaryView;  
CREATE VIEW AnnualSalaryView AS  
SELECT emp_id, emp_name, salary AS MonthlySalary, salary * 12 AS AnnualSalary  
FROM employees;  
SELECT * FROM AnnualSalaryView;
```

-- View 10: Update Data Through a Simple View

```
DROP VIEW IF EXISTS EmployeeSalaryView;  
CREATE VIEW EmployeeSalaryView AS  
SELECT emp_id, emp_name, salary FROM employees;  
UPDATE EmployeeSalaryView SET salary = 60000 WHERE emp_id = 101;  
SELECT * FROM EmployeeSalaryView;  
SELECT * FROM employees WHERE emp_id = 101;
```

-- View 11: CREATE OR REPLACE VIEW

```
CREATE OR REPLACE VIEW EmployeeBasicView AS  
SELECT emp_name, salary FROM employees;
```

```
SELECT * FROM EmployeeBasicView;
```

-- View 12: View Definition

```
SHOW CREATE VIEW EmployeeDepartmentView;
```

-- View 13: List All Views

```
SHOW FULL TABLES WHERE Table_type = 'VIEW';
```

-- View 14: Drop a View

```
DROP VIEW IF EXISTS SalaryRangeView;
```

```
-- =====  
-- 3. STUDENT PRACTICE TASKS SOLUTIONS (1 - 12)  
-- =====
```

-- Task 1: Create a View showing only emp_id, emp_name and salary.

```
DROP VIEW IF EXISTS Task1_EmpDetails;  
CREATE VIEW Task1_EmpDetails AS  
SELECT emp_id, emp_name, salary FROM employees;  
SELECT * FROM Task1_EmpDetails;
```

-- Task 2: Create a View for employees earning more than 60000.

```
DROP VIEW IF EXISTS Task2_HighEarnings;  
CREATE VIEW Task2_HighEarnings AS  
SELECT * FROM employees WHERE salary > 60000;  
SELECT * FROM Task2_HighEarnings;
```

-- Task 3: Create a View showing employee name, department name and salary using JOIN.

```
DROP VIEW IF EXISTS Task3_EmpDept;  
CREATE VIEW Task3_EmpDept AS  
SELECT e.emp_name, d.dept_name, e.salary  
FROM employees e  
JOIN departments d ON e.dept_id = d.dept_id;  
SELECT * FROM Task3_EmpDept;
```

-- Task 4: Create a View showing only ECE employees.

```
DROP VIEW IF EXISTS Task4_ECE_Employees;  
CREATE VIEW Task4_ECE_Employees AS  
SELECT e.emp_id, e.emp_name, d.dept_name, e.salary  
FROM employees e  
JOIN departments d ON e.dept_id = d.dept_id  
WHERE d.dept_name = 'ECE';  
SELECT * FROM Task4_ECE_Employees;
```

-- Task 5: Create a View showing department-wise employee count.

```
DROP VIEW IF EXISTS Task5_DeptEmpCount;  
CREATE VIEW Task5_DeptEmpCount AS  
SELECT d.dept_name, COUNT(e.emp_id) AS TotalEmployees  
FROM employees e  
JOIN departments d ON e.dept_id = d.dept_id  
GROUP BY d.dept_id, d.dept_name;  
SELECT * FROM Task5_DeptEmpCount;
```

-- Task 6: Create a View showing department-wise maximum salary.

```
DROP VIEW IF EXISTS Task6_DeptMaxSalary;  
CREATE VIEW Task6_DeptMaxSalary AS  
SELECT d.dept_name, MAX(e.salary) AS MaxSalary  
FROM employees e  
JOIN departments d ON e.dept_id = d.dept_id
```

```
GROUP BY d.dept_id, d.dept_name;
SELECT * FROM Task6_DeptMaxSalary;
```

-- Task 7: Create a View showing departments whose average salary is greater than 50000.

```
DROP VIEW IF EXISTS Task7_HighAvgSalDepts;
CREATE VIEW Task7_HighAvgSalDepts AS
SELECT d.dept_name, AVG(e.salary) AS AvgSalary
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_id, d.dept_name
HAVING AVG(e.salary) > 50000;
SELECT * FROM Task7_HighAvgSalDepts;
```

-- Task 8: Create a View with employee name, salary and annual salary.

```
DROP VIEW IF EXISTS Task8_AnnualSalary;
CREATE VIEW Task8_AnnualSalary AS
SELECT emp_name, salary AS MonthlySalary, (salary * 12) AS AnnualSalary
FROM employees;
SELECT * FROM Task8_AnnualSalary;
```

-- Task 9: Update an employee's salary through a simple View and verify the base table.

```
DROP VIEW IF EXISTS Task9_UpdateView;
CREATE VIEW Task9_UpdateView AS SELECT emp_id, emp_name, salary FROM employees;
UPDATE Task9_UpdateView SET salary = 52000 WHERE emp_id = 105;
SELECT * FROM employees WHERE emp_id = 105;
```

-- Task 10: Use SHOW CREATE VIEW to display the definition of a View.

```
SHOW CREATE VIEW Task3_EmpDept;
```

-- Task 11: List all Views in the database.

```
SHOW FULL TABLES WHERE Table_type = 'VIEW';
```

-- Task 12: Drop a View and verify that it no longer appears in the list.

```
DROP VIEW IF EXISTS Task1_EmpDetails;
SHOW FULL TABLES WHERE Table_type = 'VIEW';
```

-- Task 2: Create a View for employees earning more than 60000.

emp_id	emp_name	dept_id	salary
102	Priya	2	65000.00
104	Sneha	3	70000.00

-- Task 3: Create a View showing employee name, department name and salary using JOIN.

emp_name	dept_name	salary
Rahul	CSE	60000.00
Priya	ECE	65000.00
Anil	CSE	55000.00
Sneha	EEE	70000.00
Kiran	ECE	52000.00

-- Task 4: Create a View showing only ECE employees.

emp_id	emp_name	dept_name	salary
102	Priya	ECE	65000.00
105	Kiran	ECE	52000.00

-- Task 5: Create a View showing department-wise employee count.

dept_name	TotalEmployees
CSE	2
ECE	2
EEE	1

-- Task 6: Create a View showing department-wise maximum salary.

dept_name	MaxSalary
CSE	60000.00
ECE	65000.00
EEE	70000.00

-- Task 7: Create a View showing departments whose average salary is greater than 50000.

dept_name	AverageSalary
CSE	57500.000000
ECE	58500.000000
EEE	70000.000000

-- Task 8: Create a View with employee name, salary and annual salary.

emp_name	MonthlySalary	AnnualSalary
Rahul	60000.00	720000.00
Priya	65000.00	780000.00
Anil	55000.00	660000.00
Sneha	70000.00	840000.00

emp_id	emp_name	dept_id	salary
105	Kiran	2	52000.00
NULL	NULL	NULL	NULL

-- Task 9: Update an employee's salary through a simple View and verify the base table.

-- Task 10: Use SHOW CREATE VIEW to display the definition of a View.

View	Create View	character_set_cli...	collation_connecti...
task3_empdept	CREATE ALGORITHM=UNDEFINED DEFINER...	utf8mb4	utf8mb4_0900_ai_ci

-- Task 11: List all Views in the database.

Tables_in_employeeedb2	Table_type
annualsalaryview	VIEW
cseemployees	VIEW
departmentsalaryview	VIEW
employeebasicview	VIEW
employeeedepartmentview	VIEW
employeesalaryview	VIEW
employeeview	VIEW
highaveragedepartments	VIEW
highsalaryemployees	VIEW
task2_highearners	VIEW
task3_empdept	VIEW
task4_ece_employees	VIEW
task5_depttempcount	VIEW
task6_deptmaxsalary	VIEW
task7_highavgsaldepts	VIEW
task8_annualsalary	VIEW
task9_updateview	VIEW

-- Task 12: Drop a View and verify that it no longer appears in the list.

Tables_in_employeeedb2	Table_type
annualsalaryview	VIEW
cseemployees	VIEW
departmentsalaryview	VIEW
employeebasicview	VIEW
employeeedepartmentview	VIEW
employeesalaryview	VIEW
employeeview	VIEW
highaveragedepartments	VIEW
highsalaryemployees	VIEW
task2_highearners	VIEW
task3_empdept	VIEW
task4_ece_employees	VIEW
task5_depttempcount	VIEW
task6_deptmaxsalary	VIEW
task7_highavgsaldepts	VIEW
task8_annualsalary	VIEW
task9_updateview	VIEW

-- View 1: Display All Employees

emp_id	emp_name	dept_id	salary
101	Rahul	1	60000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	52000.00

-- View 2: Selected Columns

emp_name	salary
Rahul	60000.00
Priya	65000.00
Anil	55000.00
Sneha	70000.00
Kiran	52000.00

-- View 3: High Salary Employees

emp_id	emp_name	salary
101	Rahul	60000.00
102	Priya	65000.00
104	Sneha	70000.00

-- View 4: Employee and Department using JOIN

emp_id	emp_name	dept_name	salary
101	Rahul	CSE	60000.00
102	Priya	ECE	65000.00
103	Anil	CSE	55000.00
104	Sneha	EEE	70000.00
105	Kiran	ECE	52000.00

-- View 5: Employees from a Particular Department

emp_id	emp_name	dept_name	salary
101	Rahul	CSE	60000.00
103	Anil	CSE	55000.00

-- View 6: Department-wise Average Salary

dept_name	EmployeeCount	AverageSalary
CSE	2	57500.000000
ECE	2	58500.000000
EEE	1	70000.000000

-- View 7: Departments with Average Salary Above 55000

dept_name	AverageSalary
CSE	57500.000000
ECE	58500.000000
EEE	70000.000000

-- View 9: View with Calculated Column

emp_id	emp_name	MonthlySalary	AnnualSalary
101	Rahul	60000.00	720000.00
102	Priya	65000.00	780000.00
103	Anil	55000.00	660000.00
104	Sneha	70000.00	840000.00
105	Kiran	52000.00	624000.00

-- View 10: Update Data Through a Simple View

emp_id	emp_name	dept_id	salary
101	Rahul	1	60000.00
NULL	NULL	NULL	NULL

-- View 11: CREATE OR REPLACE VIEW

emp_name	salary
Rahul	60000.00
Priya	65000.00
Anil	55000.00
Sneha	70000.00
Kiran	52000.00

-- View 12: View Definition

View	Create View	character_set_cli...	collation_connecti...
employee departmentview	CREATE ALGORITHM=UNDEFINED DEFINER...	utf8mb4	utf8mb4_0900_ai_ci

-- View 13: List All Views

Tables_in_employeeedb2	Table_type	
annualsalaryview	VIEW	
cseemployees	VIEW	
departmentsalaryview	VIEW	
employeebasicview	VIEW	
employeeedepartmentview	VIEW	
employeesalaryview	VIEW	
employeeview	VIEW	
highaveragedepartments	VIEW	
highsalaryemployees	VIEW	
task2_highearners	VIEW	
task3_empdept	VIEW	
task4_ece_employees	VIEW	
task5_depttempcount	VIEW	
task6_deptmaxsalary	VIEW	
task7_highavgсалдеpts	VIEW	
task8_annualsalary	VIEW	
task9_updateview	VIEW	